Review 1: Hospital Emergency Resource Manager (HERM)

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1. Problem Statement & Motivation

Hospital Emergency Departments frequently face unpredictable surges of critical patients competing for scarce medical resources (ICU beds, ventilators, trauma physicians, nurses, and operating theatres). When handled manually or through passive software, three severe failures occur: wrong ordering (priority inversion where stable cases block critical emergencies), conflicting requests (race conditions causing double-allocations), and deadlocks (patients holding resources while waiting indefinitely for others).

This project addresses these challenges by mapping the emergency department directly onto an Operating System. Patients are abstracted as computational processes, and scarce medical assets are treated as lockable, preemptible system devices governed by OS priority scheduling, counting semaphores, Banker's algorithm, and Wait-For Graph deadlock detection.

2. Project Objectives

1. Model emergency patients as processes with priority, arrival time, treatment burst, and multi-resource requirements.

2. Implement hybrid preemptive priority scheduling with aging for critical cases, and Round Robin for non-critical cases.

3. Coordinate access to limited resources with counting semaphores and mutex locks to eliminate race conditions.

4. Apply Banker's algorithm for deadlock avoidance, and runtime Wait-For Graph DFS cycle detection with preemption-based recovery.

5. Provide an aesthetic real-time command dashboard with IPC event streams, and compare FCFS, Non-Preemptive, and Preemptive Priority with Aging.

3. OS Concept to Emergency Department Mapping

Operating System Concept	Hospital Implementation
Process	Patient emergency request (arrival time, priority level, burst time, resource demands)
Preemptive Scheduling	Newly arrived Level 1 Critical patients immediately pause running lower-priority tasks
Priority Aging	Waiting patient priority is elevated every Ta ticks (capped at Level 2) to eliminate starvation
Counting Semaphore	Controls available units of ICU Beds (4), Ventilators (3), Doctors (5), Nurses (8), Theatres (2)
Allocation Mutex	Serializes state checks and allocations into an atomic step, preventing double-booking
Banker's Algorithm	Tests hypothetical grant state safety to proactively avoid deadlock conditions
Wait-For Graph (WFG)	Directed graph tracking resource dependencies; DFS finds cycles and preempts victim
IPC Bus	Message queue delivering discrete ARRIVAL, REQUEST, GRANT, PREEMPT, and DONE events

4. Literature Review (16 Authentic Google Scholar Citations)

A comprehensive literature survey was conducted across IEEE, ACM, Springer, Elsevier, INFORMS, and Wiley to evaluate state-of-the-art triage and scheduling approaches:

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[6] A. Silberschatz, P. B. Galvin, and G. Gagne, Operating System Concepts, 10th ed. Hoboken, NJ: John Wiley & Sons, 2018.

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[9] E. W. Dijkstra, 'Cooperating sequential processes,' Technische Hogeschool Eindhoven, Tech. Rep. EWD-123, 1965.

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[14] R. C. Holt, 'Some deadlock properties of computer systems,' ACM Comput. Surv., vol. 4, no. 3, pp. 179-196, 1972.

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[16] M. Armony, S. Israelit, A. Mandelbaum, et al., 'On patient flow in hospitals: A data-based queueing-science perspective,' Stoch. Syst., vol. 5, no. 1, pp. 146-194, 2015.